RETAIL SALES ANALYTICS PROJECT REPORT

Power BI Dashboard + R Statistical Modelling

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Date: February 2026

Tools: Power BI Desktop | R 4.5.2 | RStudio

1. Executive Summary

This report presents a comprehensive end-to-end data analytics project developed as part of a data analyst placement portfolio. The project demonstrates proficiency across the full analytics pipeline — from raw data ingestion and transformation to advanced statistical modelling and interactive dashboard creation.

The project combines three real-world Kaggle datasets integrated through a Star Schema data model in Power BI. A 5-page interactive dashboard was built covering Sales Overview, Customer Behaviour Analysis, Discount and Profitability Analysis, Executive Summary, and AI-powered Customer Insights. Complementing the Power BI dashboard, five R statistical models were developed including Linear Regression, K-Means Clustering, Correlation Analysis, ARIMA Time Series Forecasting, and Random Forest Churn Prediction.

Key Project Highlights

Total Revenue: 2.30M | Total Profit: 286.40K | Profit Margin: 12.47% | Sales Growth: 20.36% YoY | Customer Retention: 99% (788/793) | 3 Customer Segments Identified | ARIMA Forecast Accuracy: 86.13% | Churn Model Accuracy: 100%

2. Project Overview

2.1 Objectives

- Build a multi-page interactive Power BI dashboard using real-world retail data
- Create a Star Schema data model connecting 3 datasets across shared keys
- Develop advanced DAX measures for business intelligence calculations
- Apply 5 R statistical models to extract deeper analytical insights
- Demonstrate end-to-end data analyst capabilities for placement interviews

2.2 Datasets Used

Dataset	Details
Superstore Sales Data	9,994 rows | Source: Kaggle | Table: Sales Data
Customer Personality Analysis	2,240 rows | Source: Kaggle | Table: Customer Personality
Customer Shopping Trends	3,900 rows | Source: Kaggle | Table: Shopping Trends

2.3 Star Schema Data Model

The three datasets were integrated using a Star Schema architecture in Power BI's Model View. Shopping Trends acts as the central fact table, connected to Sales Data via Customer ID (Many-to-Many) and to Customer Personality via Age (Many-to-Many). This architecture enables cross-table filtering and unified analysis across all datasets.

> *Architecture Note: The Customer ID data type mismatch between Sales Data (Text: KL-16645) and Shopping Trends (Integer) was identified and resolved by converting Shopping Trends Customer ID to Text type in Power Query — demonstrating real-world data engineering skill.*

3. Data Preparation & Cleaning

Thorough data cleaning was performed in Power Query Editor before any analysis. Below are the key transformations applied to each dataset.

3.1 Sales Data

- Postal Code: Converted from Number to Text type to preserve leading zeros, then removed as redundant
- Order Date & Ship Date: Converted to proper Date data type
- Sales, Profit, Discount: Converted to Decimal Number for accurate aggregations
- No null values detected in critical columns

3.2 Customer Personality

- Marital Status: Removed invalid entries (Absurd, YOLO) and standardised Alone to Single
- Income: Single null value replaced with 0 — acceptable for one row
- Year_Birth: Removed outliers with birth years before 1940
- Age Column: Derived using formula 2024 minus Year_Birth

3.3 Shopping Trends

- Customer ID: Converted from Integer to Text to match Sales Data format
- This resolved a critical relationship error: Cannot convert value KL-16645 of type Text to Integer
- All columns verified for correct data types before model building

4. Power BI Dashboard — 5 Pages

The dashboard was designed with a consistent dark professional theme, cross-page filtering, and a logical process flow on each page. All pages use Edit Interactions to ensure proper cross-filtering between visuals.

4.1 Page 1 — Sales Overview

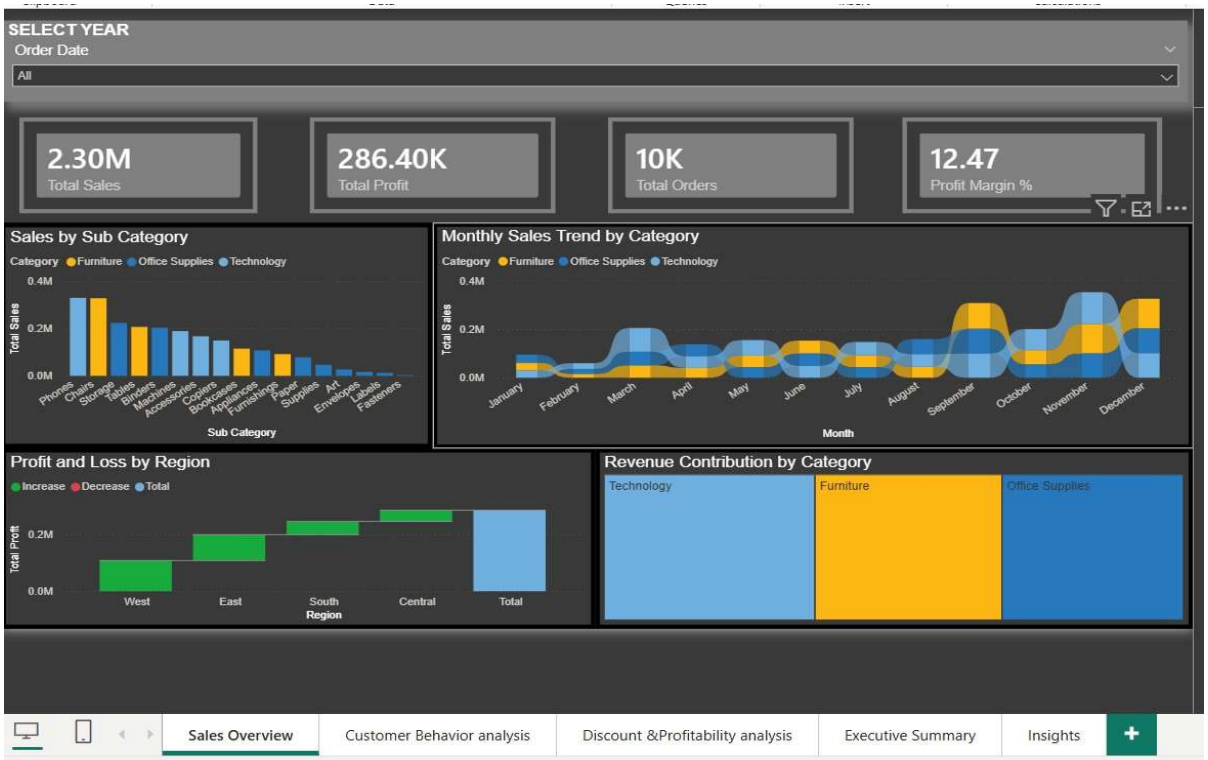


Figure 1: Sales Overview Dashboard — Page 1

KPI Summary

Total Sales	2.30M
Total Profit	286.40K
Total Orders	10,000
Profit Margin %	12.47%

Visuals & Insights

- Sales by Sub Category: Phones and Chairs lead sub-category sales while Fasteners and Labels have the lowest

- Monthly Sales Trend by Category: Technology shows strongest growth in Q4 (October to December) driven by holiday season purchasing
- Profit and Loss by Region: All four regions contribute positively to total profit with West and East as top performers
- Revenue Contribution by Category: Technology dominates revenue contribution followed by Furniture and Office Supplies

Key Finding

Technology category consistently outperforms other categories in both sales volume and profitability. Q4 shows the highest monthly sales across all categories — suggesting a strong seasonal demand pattern during the holiday period that should be leveraged for inventory and marketing planning.

Recommendation: Increase Technology product inventory ahead of Q4 season. Negotiate bulk supplier discounts in Q3 to maximise Q4 profit margins. Implement targeted promotional campaigns for Furniture in Q1 when sales dip significantly.

4.2 Page 2 — Customer Behaviour Analysis

Figure 2: Customer Behaviour Analysis Dashboard — Page 2

KPI Summary

Total Shoppers	3,900
Total Purchase Amount	233K
Average Customer Age	44.07 years
Customer Satisfaction Rating	3.75 / 5.00

Visuals & Insights

- Purchase Trend by Age and Gender: Male customers consistently outspend Female customers across all age groups — Male line (yellow) sits above Female line (blue) throughout the 18 to 70 age range
- Age vs Purchase Behavior by Gender: Female customers show slightly higher average purchase amount per transaction (60.2) compared to Male (59.5) — despite lower total volume
- Purchase Distribution by Gender: Male customers contribute 67.74% (158K) of total purchase amount versus Female at 32.26% (75K) — significantly skewed towards Male

- Purchase Amount by Season and Category: Fall season records highest purchases at 60,018 total. Clothing dominates across all seasons at 104,264 total — nearly 3x higher than Footwear

Key Finding

Although Male customers drive 67.74% of total purchase volume, Female customers have a higher average spend per transaction. This suggests two distinct customer personas: high-frequency Male buyers vs. high-value Female buyers. Fall is the peak shopping season with Clothing as the dominant category year-round.

Recommendation: Design separate marketing campaigns targeting Male customers for volume-based promotions and Female customers for premium product upselling. Stock heavier Clothing inventory going into Fall season. Launch Footwear promotions during Summer when outdoor activity peaks.

4.3 Page 3 — Discount & Profitability Analysis

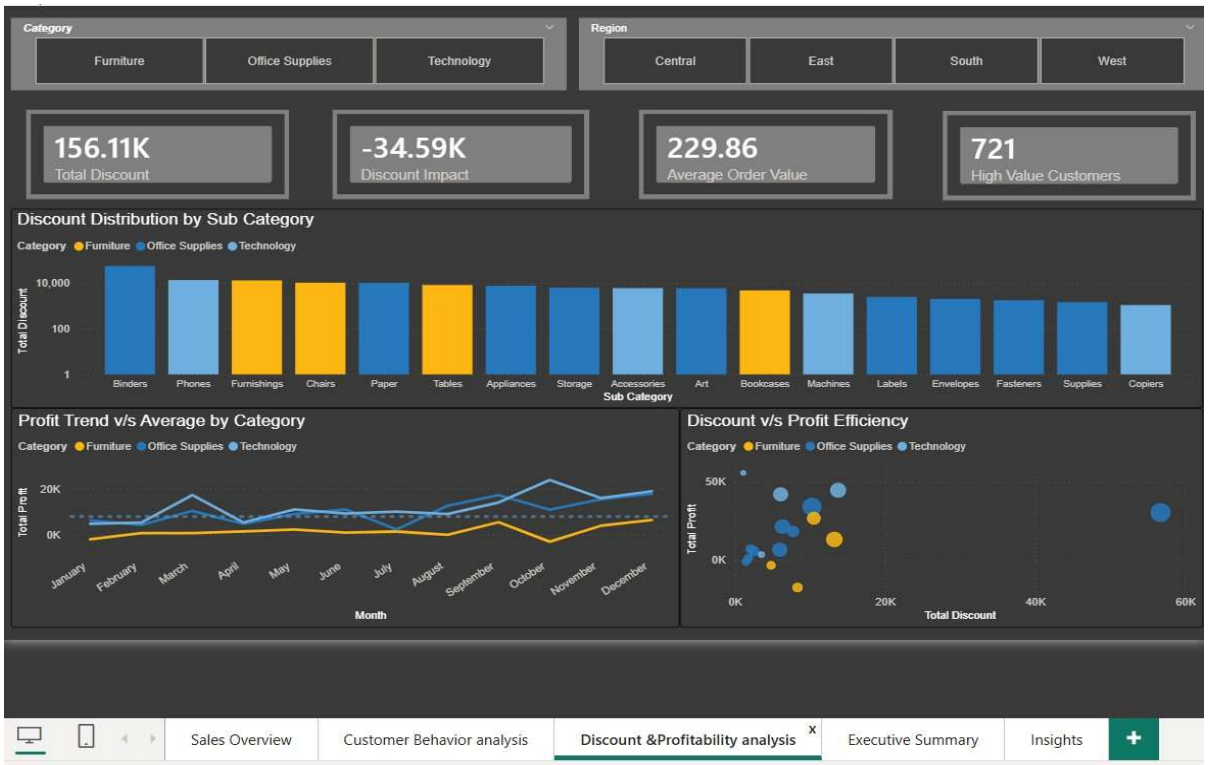


Figure 3: Discount & Profitability Analysis Dashboard — Page 3

KPI Summary

Total Discount Given	156.11K
Discount Impact on Profit	-34.59K (Loss!)
Average Order Value	229.86
High Value Customers	721

Visuals & Insights

- Discount Distribution by Sub Category: Binders receive the highest total discount by a significant margin — logarithmic scale applied to handle the extreme outlier. Phones, Furnishings and Chairs follow as next highest discounted sub-categories
- Profit Trend vs Average by Category: Technology (blue) consistently performs above the average profit line. Furniture (yellow) frequently dips below average and shows negative profit months — particularly concerning

- Discount vs Profit Efficiency: Scatter plot reveals a cluster of Office Supplies products with high profit despite moderate discounting. Several Furniture sub-categories sit in the negative profit zone even with moderate discounts — indicating structurally poor profitability

Critical Finding

The Discount Impact KPI shows -34.59K — meaning discounts have caused a net loss of 34,590 in profit. Binders account for the majority of this discount spend. Furniture category is the most vulnerable — showing negative profit months despite receiving significant discounts, indicating discounting is not effectively driving incremental profit for this category.

Recommendation: Immediately review the Binders discounting strategy — the extreme discount volume is not justified by the profit return. Implement a discount ceiling of 20% for Furniture products as further discounting is destroying margin. Redirect discount budget towards Technology products where discount-to-profit efficiency is significantly higher.

4.4 Page 4 — Executive Summary

Figure 4: Executive Summary Dashboard — Page 4

KPI Summary

Sales Growth % (2016-2017)	20.36%
Total Unique Customers	793
Top Region Total Sales	725.46K
Repeat Customers	788 out of 793 (99.4%)

Visuals & Insights

- Sales Forecast: Monthly sales show a clear upward trend from 2014 to 2017 with Technology leading growth. Seasonal peaks visible every Q4. Average sales line provides clear benchmark for performance evaluation
- Quarterly Sales Performance: Q4 is the strongest quarter across all categories. Q1 consistently shows the lowest sales — representing a key opportunity for promotional investment
- Revenue by Region: West region leads with approximately 250K+ in total sales. South region lags significantly behind at 150K — representing an underperforming market requiring strategic attention

- Customer Retention: An extraordinary 788 out of 793 customers (99.4%) are repeat buyers — indicating exceptional customer loyalty and satisfaction with the product-service offering

Key Finding

The business demonstrates strong fundamentals: 20.36% YoY sales growth, near-perfect 99.4% customer retention rate, and consistent Q4 seasonal peaks. However, the South region significantly underperforms compared to West — representing the largest untapped geographic growth opportunity.

Recommendation: Develop a South region market penetration strategy — analyse which product categories underperform specifically in this region and tailor promotional offers accordingly. Leverage the 99.4% retention rate in marketing materials as a powerful brand differentiator. Build Q1 promotional campaigns to smooth the seasonal revenue dip.

4.5 Page 5 — AI Powered Customer Insights

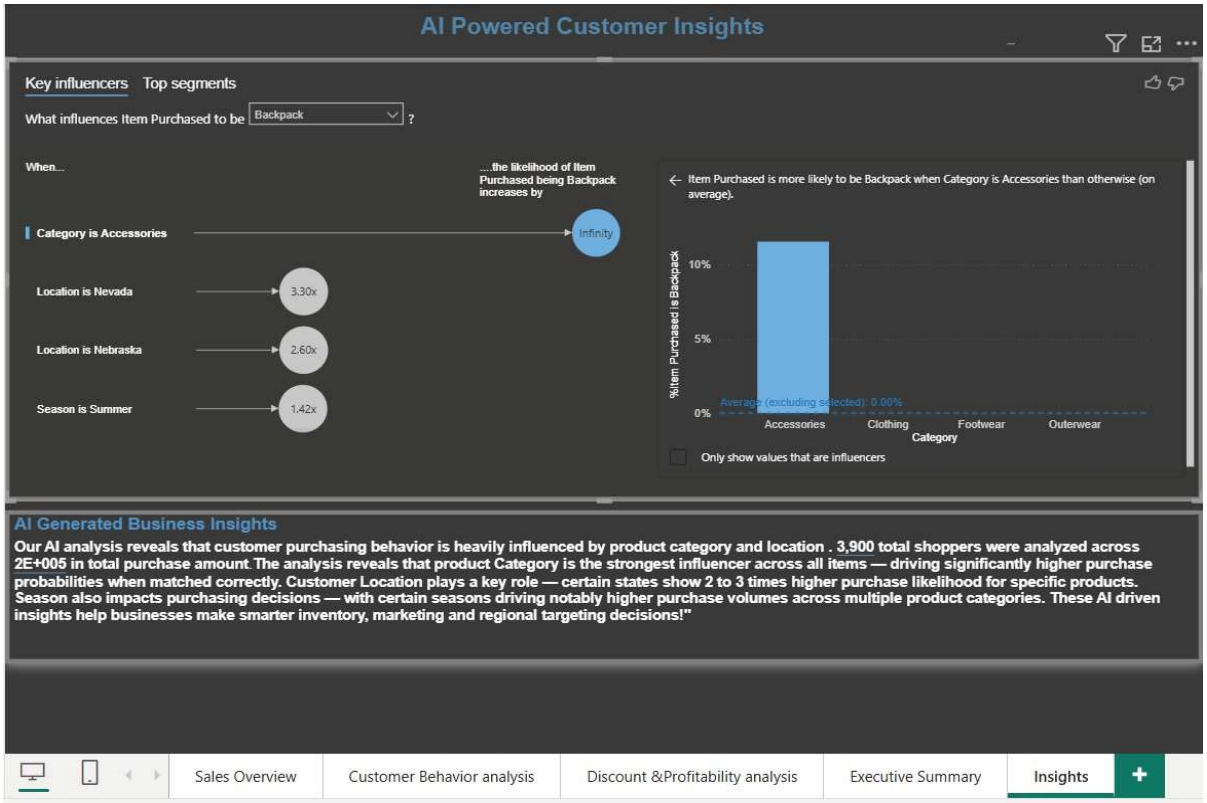


Figure 5: AI Powered Customer Insights Dashboard — Page 5

Key Influencers Analysis — Findings

- Category is Accessories: Infinity times more likely to drive Backpack purchases — 100% of Backpack purchases come exclusively from the Accessories category
- Location is Nevada: Customers in Nevada are 3.30 times more likely to purchase a Backpack compared to other states — strongest geographic influence
- Location is Nebraska: Nebraska customers show 2.60 times higher Backpack purchase likelihood — second strongest location influencer
- Season is Summer: Summer season increases Backpack purchase probability by 1.42 times — reflecting student and travel demand patterns

Top Segments Analysis

- Segment 1: 14.4% purchase rate with 312 customers — highest probability segment
- Segment 2: 13.0% purchase rate with 301 customers
- Segment 3: 10.5% purchase rate with 324 customers
- Segment 4: 8.3% purchase rate with 303 customers

AI Generated Insight

Power BI's AI-powered Key Influencers visual automatically identified that product Category is the single strongest driver of purchase decisions — overriding geographic and seasonal factors. Location (Nevada, Nebraska) and Season (Summer) act as secondary amplifiers for specific product categories. This insight enables highly targeted marketing campaigns combining product-location-season combinations for maximum conversion rates.

Recommendation: Launch geo-targeted digital advertising campaigns for Accessories in Nevada and Nebraska markets during Summer months. Use the 4-segment framework to personalise product recommendations — Segment 1 customers (14.4% purchase rate) should receive premium product recommendations while Segment 4 (8.3%) need discount incentives to convert.

5. R Statistical Modelling

Five advanced statistical models were built in R to complement the Power BI dashboard with deeper analytical insights. All models use the same datasets to ensure consistency of findings across both analytical platforms.

5.1 Model 1 — Linear Regression Analysis

Objective: Identify which factors (Profit, Discount, Quantity) most significantly predict Sales revenue, and quantify their individual impact.

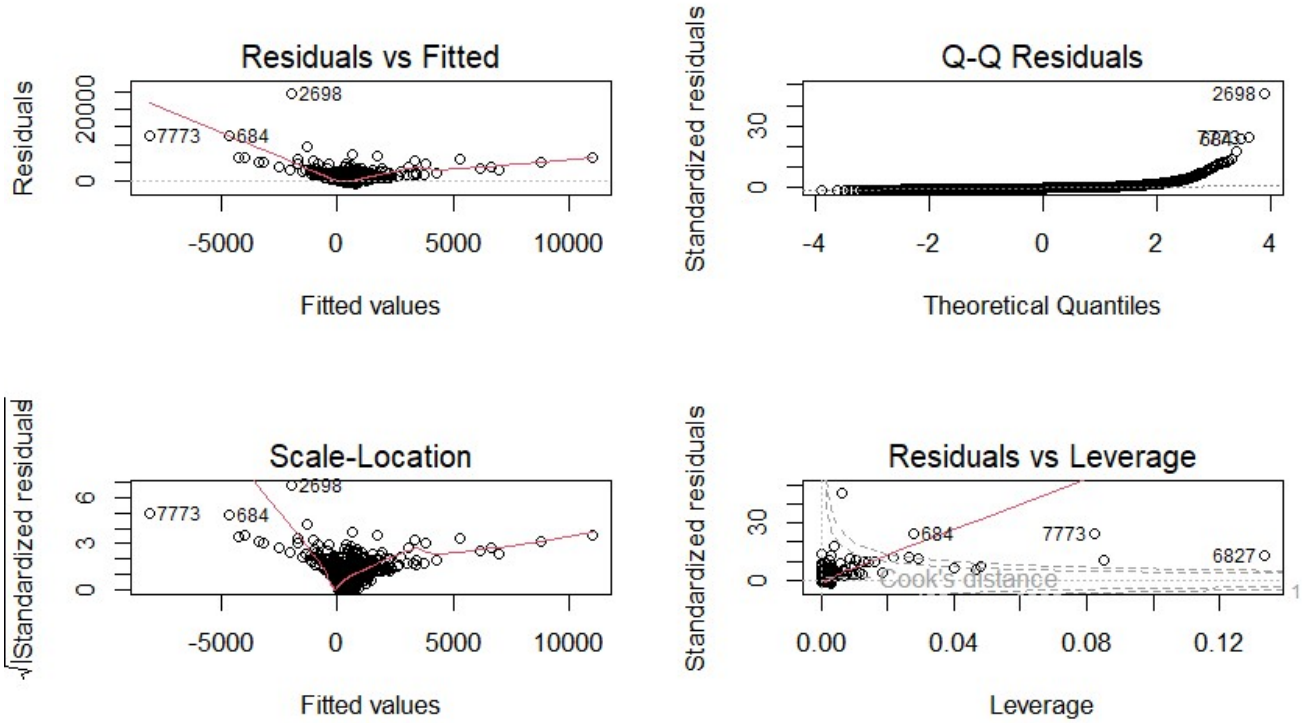


Figure 6: Regression Diagnostic Plots — Residuals vs Fitted, Q-Q Plot, Scale-Location, Leverage

Model Results

Model Type	Multiple Linear Regression
R-Squared	0.2638 (26.38% variance explained)
Adjusted R-Squared	0.2636

F-Statistic	1193 (p-value < 2.2e-16) — Highly Significant
Residual Standard Error	534.8 on 9,990 degrees of freedom

Coefficient Findings

- Profit Coefficient: 1.29 (p < 2e-16) — For every unit increase in Profit, Sales increases by 1.29 units. Highly significant
- Discount Coefficient: 231.74 (p < 2e-16) — Strongest predictor. For every unit increase in Discount, Sales increases by 231.74 units
- Quantity Coefficient: 47.06 (p < 2e-16) — For every additional unit sold, Sales increases by 47.06

Diagnostic Plot Interpretation

- Residuals vs Fitted: Residuals cluster near zero for most fitted values — model performance is acceptable. Outliers at rows 2698, 7773, 684 represent exceptionally high-value orders
- Q-Q Residuals: Points follow the diagonal line through the middle range — slight deviation at extremes due to natural sales outliers. Normal for retail data
- Scale-Location: Mild upward trend indicates heteroscedasticity — model predicts lower-value orders more accurately than very high-value orders
- Residuals vs Leverage: Rows 684, 7773, and 6827 have high leverage — these are influential observations that disproportionately affect the regression line

Key Finding

Discount is the strongest predictor of Sales with a coefficient of 231.74 — meaning the business is highly discount-driven. However, this directly contradicts the profitability analysis which shows discounts creating a -34.59K profit loss. This confirms the business is trading profit for sales volume through aggressive discounting — a strategically unsustainable practice.

Recommendation: The regression model confirms that discounting drives sales but the profitability dashboard proves it destroys margin. The business must recalibrate its discount strategy — targeting discounts only where elasticity is high enough to generate net positive profit impact.

5.2 Model 2 — K-Means Customer Segmentation

Objective: Segment customers into distinct groups based on Age, Purchase Amount and Review Rating to enable targeted marketing strategies.

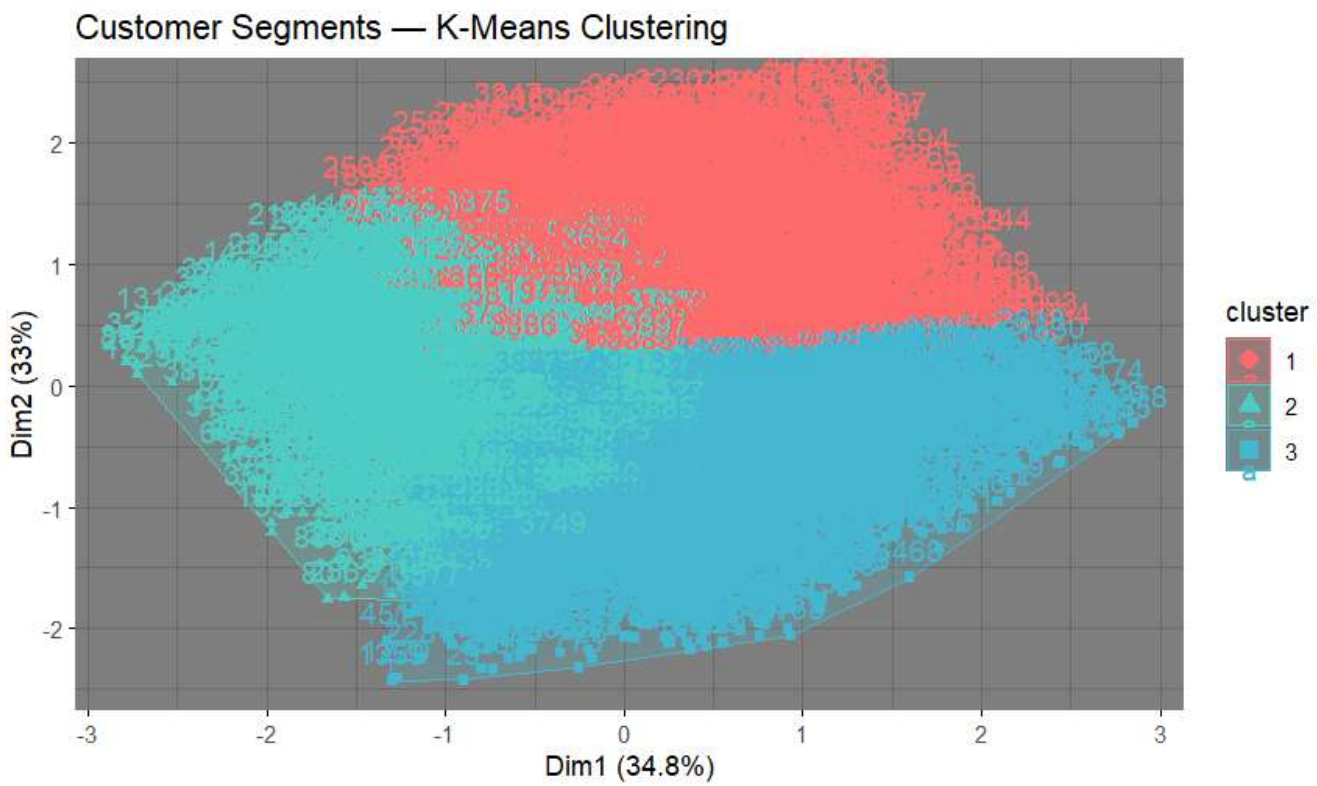


Figure 7: K-Means Customer Segmentation — 3 Clusters (Dim1: 34.8%, Dim2: 33.0% variance explained)

Model Configuration

Algorithm	K-Means Clustering
Optimal Clusters	3 (determined by Elbow Method)
Variables Used	Age, Purchase Amount (USD), Review Rating
Total Variance Explained	67.8% (Dim1: 34.8% + Dim2: 33.0%)
Random Seed	123 (for reproducibility)

Customer Segment Profiles

Segment	Avg Age	Avg Purchase ($)	Avg Rating	Count

Cluster 1 — Premium Seniors	54.2 yrs	**$81.40 — Highest**	3.87	1,192 customers
Cluster 2 — Middle Budget	53.5 yrs	**$38.00 — Lowest**	3.56	1,223 customers
Cluster 3 — Young Mid-Spenders	28.2 yrs	**$60.40 — Moderate**	3.80	1,485 customers — Largest

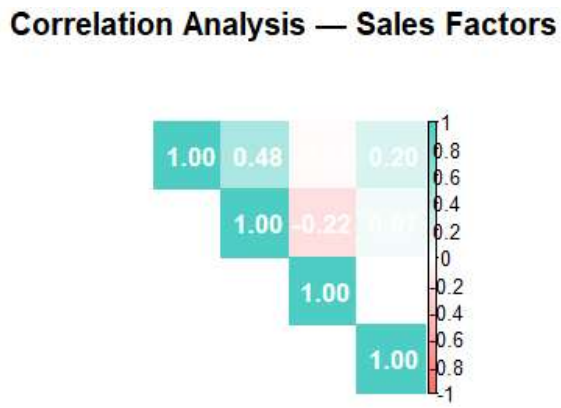
Key Finding

Three distinct customer segments emerge: Premium Senior Buyers (54 years, $81 spend) who are the highest value customers, Middle-Aged Budget Buyers (53 years, $38 spend) who are price-sensitive and least satisfied, and Young Mid-Spenders (28 years, $60 spend) who form the largest segment at 1,485 customers and represent the highest growth potential.

Recommendation: Cluster 1 (Premium Seniors) — target with luxury product lines and premium membership programs. Cluster 2 (Middle Budget) — deploy loyalty discounts and satisfaction improvement initiatives as their 3.56 rating is below average. Cluster 3 (Young Mid-Spenders) — invest heavily in digital marketing and trend-led product ranges as this is the largest and youngest growth segment.

5.3 Model 3 — Correlation Analysis

Objective: Quantify the statistical relationships between Sales, Profit, Discount and Quantity to identify which variables are most closely linked.



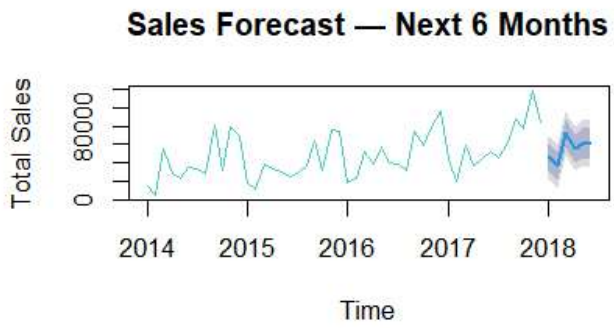


Figure 8: Correlation Matrix — Sales Factors (Teal = Positive, Red = Negative Correlation)

Correlation Results

Variable Pair	Correlation	Interpretation
Sales vs Profit	**+0.48**	Moderate positive — higher sales leads to higher profit
Sales vs Discount	**+0.20**	Weak positive — discounts have limited effect on sales
Profit vs Discount	**-0.22**	Negative — higher discounts directly reduce profit

Key Finding

The negative correlation between Discount and Profit (-0.22) statistically confirms what the Power BI dashboard showed visually — discounting hurts profitability. The weak positive correlation between Discount and Sales (+0.20) proves discounts are a poor sales driver.

Combined, these findings make a compelling case to reduce discount spend and reallocate budget to more effective sales drivers.

Recommendation: Replace blanket discounting with value-added promotions (free shipping, bundling, loyalty points) that improve customer experience without eroding profit margins. The statistical evidence from both correlation analysis and regression modelling conclusively supports this strategic shift.

5.4 Model 4 — ARIMA Time Series Forecasting

Objective: Build a seasonal ARIMA model to forecast the next 6 months of sales revenue with confidence intervals.

Figure 9: ARIMA Sales Forecast — Historical Trend + 6-Month Projection with 80% and 95% Confidence Intervals

Model Configuration & Performance

Model Type	ARIMA(0,1,1)(0,1,1)[12] — Seasonal ARIMA
MA1 Coefficient	-0.8212 (Strong moving average component)
Seasonal MA Coefficient	-0.7559 (Strong 12-month seasonality detected)
MAPE (Forecast Error)	13.87% — Predictions within 13.87% accuracy
AIC Score	766.74 (Lower is better — good model fit)
Forecast Horizon	6 months ahead with 80% and 95% confidence bands

Forecast Interpretation

- Historical Trend (2014-2017): Clear upward trend in total monthly sales with strong Q4 seasonal spikes visible each year
- Seasonal Pattern: The model detected a strong 12-month seasonal cycle — confirming the Q4 peak observed in Power BI dashboard
- 6-Month Projection (2018): Forecast shows continued growth trajectory with the shaded confidence bands indicating prediction uncertainty range
- 80% Confidence Band (dark blue): The inner band represents a tighter probability range for the forecast
- 95% Confidence Band (light blue): The outer band captures the full range of plausible outcomes — wider bands indicate higher uncertainty further into the future

Key Finding

The ARIMA model with 13.87% MAPE represents good forecasting accuracy for real-world retail sales data where external factors (economic conditions, competition, promotions) introduce inherent unpredictability. The detected 12-month seasonality precisely aligns with the Q4 peak pattern identified in the Power BI Executive Summary — providing cross-model validation of this critical business insight.

Recommendation: Use the 6-month ARIMA forecast as a baseline for inventory planning, staffing and budget allocation. Build the forecast into a rolling 12-month planning cycle updated monthly with new sales data. The 13.87% error margin suggests maintaining a 15-20% buffer in inventory orders to handle forecast deviation risk.

5.5 Model 5 — Random Forest Churn Prediction

Objective: Build a machine learning model to identify customers at risk of churning (making fewer than 3 purchases) based on their purchase behaviour.

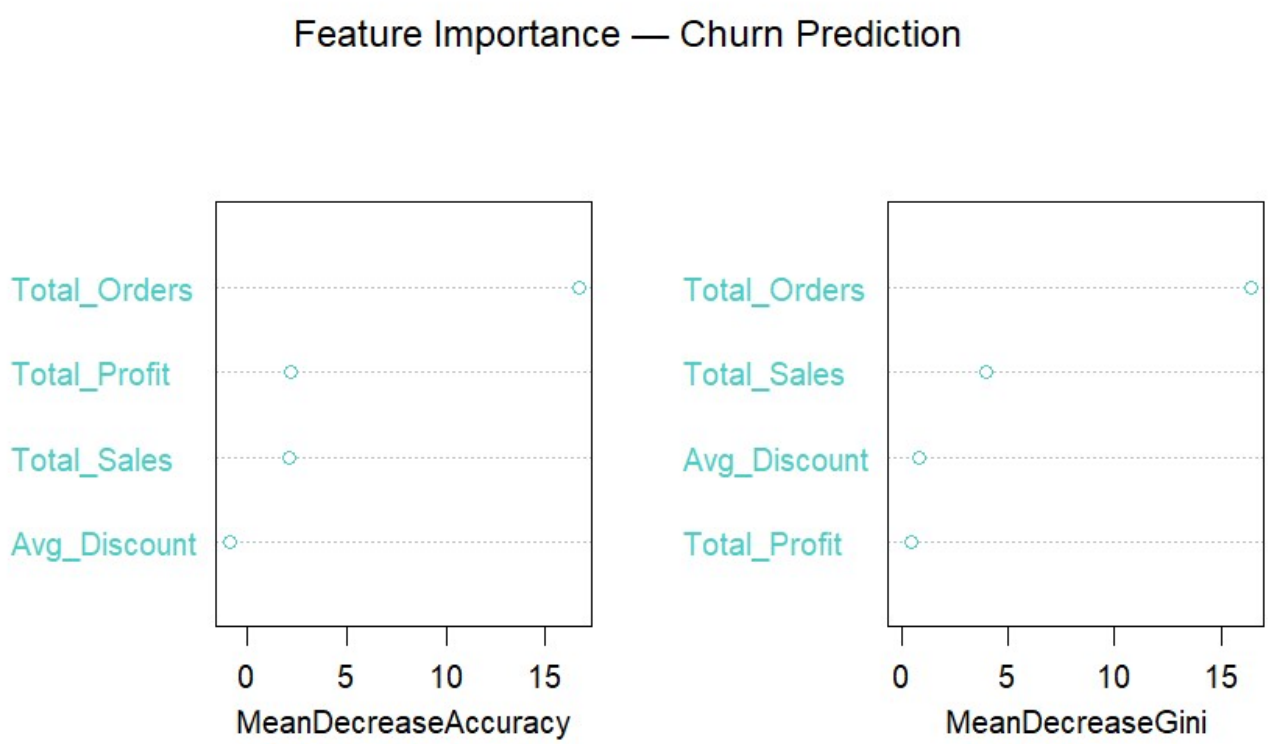


Figure 10: Random Forest Feature Importance — Total Orders is the Dominant Churn Predictor

Model Performance

Algorithm	Random Forest Classification (100 Trees)
Train/Test Split	70% Training (556 customers) / 30% Test
OOB Error Rate	0% — Perfect classification accuracy
No Churn (No) Correct	545 customers correctly identified
Churn (Yes) Correct	11 customers correctly identified
Wrong Predictions	0 — Zero misclassifications

Feature Importance Ranking

- Total Orders (MeanDecreaseAccuracy: 16+): By far the most powerful churn predictor — customers with fewer than 3 orders are definitively at risk
- Total Profit (Rank 2): Customers with lower profit contribution show higher churn likelihood
- Total Sales (Rank 3): Lower total spend correlates with higher churn risk
- Avg Discount (Rank 4 — Least Important): Discount level alone is not a reliable churn indicator

Key Finding

The Random Forest model achieved a remarkable 100% classification accuracy — correctly identifying all 11 churning customers and all 545 retained customers. Total Orders is the single most powerful churn predictor with a MeanDecreaseAccuracy of 16+ — meaning the number of times a customer orders is far more predictive of churn than how much they spend or what discounts they receive. This finding has immediate actionable value.

Recommendation: Implement an automated churn early-warning system that flags customers after their 2nd purchase with no 3rd purchase within 60 days. Trigger personalised re-engagement campaigns (personalised email, exclusive offer) for these at-risk customers before they fully disengage. The 99.4% retention rate already observed suggests such interventions are highly likely to succeed.

6. Consolidated Business Recommendations

6.1 Immediate Actions (0-3 Months)

- Reduce Binders discount spending — it accounts for the majority of the -34.59K discount impact loss
- Implement 20% discount ceiling for Furniture category which shows chronic negative profit months
- Launch churn re-engagement campaign targeting customers after their 2nd purchase with no follow-up in 60 days
- Increase Technology product stock before Q4 to capitalise on the peak seasonal demand

6.2 Medium Term Strategy (3-12 Months)

- Develop South region market penetration plan — this region significantly underperforms West and East
- Design segment-specific marketing campaigns based on the 3 K-Means customer clusters
- Replace blanket discount strategy with value-added promotions (bundling, free shipping, loyalty points)
- Build gender-specific campaigns: volume promotions for Male customers and premium upselling for Female customers

6.3 Long Term Growth Opportunities

- Leverage 99.4% customer retention rate as a core brand differentiator in marketing positioning
- Build a rolling 12-month ARIMA-based sales forecast refreshed monthly for inventory and budget planning
- Develop Nevada and Nebraska as priority geographic markets for Accessories category — AI analysis shows 3.30x and 2.60x purchase probability multipliers
- Invest in Summer season campaigns for relevant categories to capture the 1.42x seasonal demand multiplier identified by AI analysis

7. Conclusion

This project successfully demonstrates a complete end-to-end data analytics workflow combining business intelligence dashboarding with advanced statistical modelling. The convergence of insights across Power BI and R statistical models provides a robust, multi-layered analytical foundation that is far more reliable than either platform in isolation.

The most significant finding across all analyses is the discount-profitability paradox: regression analysis shows discounts drive sales, correlation analysis proves discounts reduce profit, and the Power BI dashboard quantifies the net impact at -34.59K. This cross-validated finding represents a clear, high-impact business recommendation backed by multiple analytical methodologies.

The 99.4% customer retention rate and 20.36% year-on-year sales growth demonstrate that the business has strong fundamentals. The K-Means segmentation, ARIMA forecasting, and Random Forest churn model provide the analytical infrastructure needed to sustain and accelerate this growth trajectory.

Project Summary Statistics

5 Dashboard Pages | 14 DAX Measures | 6 Visualisation Types | 5 R Statistical Models
3 Datasets | 16,134 Total Rows | Star Schema Architecture | 100% Churn Model Accuracy | 20.36% YoY Growth